

The compact E_7 sigma-MGF on V_{56}

Proof and dimension-aware verification strategy

Computational proof companion

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Abstract

For simply connected compact E_7 on its minuscule 56, we prove the Molien–Weyl expression and parity rule, exactly verify the full quadratic expansion with explicit torus matrices, and state the degree-four expansion using cited invariant theory. We also give a sparse quartic verification algorithm; the present artifact does not report that computation as executed.

1 Formula and parity

The Cauchy identity and Haar projection prove

$$\mathcal{S}_{K,V} = \int_K \prod_a \det(I - t_a \rho(g))^{-1} dg = \sum_{\lambda} \dim(\mathbb{S}_{\lambda} V)^K s_{\lambda}(\mathbf{t}).$$

The equivalent monomial coefficient is $[m_{\tau}] \mathcal{S}_{K,V} = \dim(\bigotimes_i \text{Sym}^{\tau_i} V)^K$. The 56 weights of V_{56} form one Weyl orbit. Hence

$$\mathcal{S}_{E_7,56} = \frac{1}{2903040} \text{CT}_z \left[D_{E_7}(z) \prod_a \prod_{\mu \in W_{\omega_7}} (1 - t_a z^{\mu})^{-1} \right].$$

The nontrivial central element acts as $-I$, so every odd-total-degree coefficient vanishes. This statement concerns simply connected E_7 ; the 56 does not descend to E_7/Z_2 .

2 Low-degree tensors

Let $\omega \in \bigwedge^2 V^*$ be the invariant symplectic form and let $q \in \text{Sym}^4 V^*$ be the Cartan–Freudenthal quartic. At degree four, the three pairwise contractions made from two copies of ω decompose under S_4 as the Schur types $(2, 2)$ and $(1, 1, 1, 1)$. The quartic q supplies type (4) . The one-copy statement $\mathbb{C}[V_{56}]^{E_7} = \mathbb{C}[q]$ and the exhaustion of degree-four tensor invariants by the symplectic pairings and q are external invariant-theory inputs; see Brown [1]. Using those inputs,

$$\boxed{\mathcal{S}_{E_7,56} = 1 + e_2 + h_4 + s_{(2,2)} + e_4 + O_{\geq 6}}, \quad \boxed{\mathcal{S}(t, 0, \dots) = \frac{1}{1 - t^4}}.$$

The one-variable specialization kills $e_2, s_{(2,2)}, e_4$ and retains the quartic generator. In particular $([m_{(2)}], [m_{(1,1)}]) = (0, 1)$, while the predicted degree-four row is

$$([m_{(4)}], [m_{(3,1)}], [m_{(2,2)}], [m_{(2,1,1)}], [m_{(1,1,1,1)}]) = (1, 1, 2, 2, 4).$$

3 Executed quadratic verification and optional quartic confirmation

The notebook generates the 126 roots and the 56-weight orbit, constructs $M(\theta) = \text{diag}(e^{i\langle \mu, \theta \rangle})$, and verifies the determinant integrand. Jacobi–Trudi characters are integrated exactly using

$$\frac{1}{|W|} \text{CT}(Df) = \text{CT} \left(f \prod_{\alpha > 0} (1 - z^{-\alpha}) \right).$$

The complete quadratic computation is

λ	$\emptyset$	(1)	(2)	(1, 1)
proposed	1	0	0	1
exact Weyl check	1	0	0	1

All entries pass. The matrix unitarity residual is 4.2×10^{-16} and the determinant/eigenvalue residual is 8.6×10^{-14} .

Status of the degree-four row. The row follows from the cited invariant-theory decomposition. The notebook output displayed here verifies only degrees at most two. A fully mechanical confirmation may be obtained as follows, but is not executed in this artifact. Do not form dense matrices on $V^{\otimes 4}$ (dimension 56^4). Instead: (i) generate the weight-sparse root-operator action on each of $\text{Sym}^4 V$, $\mathbb{S}_{(2,2)} V$, and $\bigwedge^4 V$; (ii) retain only zero-weight basis vectors; (iii) solve the sparse raising-operator nullspace; and (iv) check dimensions 1, 1, 1. Equivalently, stream the three Schur characters through the same exact Weyl functional. An executed result must report the zero-weight dimensions, exact raising-kernel ranks, arithmetic mode, and output hash.

Reproducibility. Run `e7_verification.py` with marimo.

References

- [1] R. B. Brown, “Groups of type E_7 ,” *J. Reine Angew. Math.* **236** (1969), 79–102.
- [2] J. F. Adams, *Lectures on Exceptional Lie Groups*, Chicago Lectures in Mathematics (1996).